

Chang He (Aaron)

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EDUCATION

University of New South Wales (UNSW)

Bachelor of Science in Mathematics, UNSW 3956

Sydney, Australia

Term 3, 2024 – Present

- **Expected graduation:** Term 3, 2028 (Honours).
- **Planned transfer:** Bachelor of Computer Science, UNSW 3779.
- **Academic standing:** WAM **85**; admitted to the UNSW Talented Students Program (TSP) in 2025.

RESEARCH INTERESTS

Quantum computing: quantum algorithms and quantum error correction, including fault-tolerant computation, stabilizer/CSS codes, and algebraic methods in finite-dimensional quantum systems.

Mathematics: algebraic methods for isomorphism problems, multilinear algebra and tensor spaces, group actions, canonical forms, and related representation-theoretic tools.

Theoretical computer science: computational complexity and algorithms, especially graph and tensor isomorphism, algebraic reductions, and structural problems at the interface of algebra and computation.

RESEARCH EXPERIENCE

Independent Study, UTS

Sydney, Australia

Quantum Computing, Algebraic Algorithms, and Isomorphism Problems with Youming Qiao *Term 2, 2025 – Present*

- Conducting supervised reading and research preparation in quantum computation, complexity theory, and algebraic methods for isomorphism problems.
- Working on graph isomorphism and tensor isomorphism problems, with emphasis on algebraic formulations, reductions, and invariants for equivalence testing.
- Preparing a manuscript on connections between graph isomorphism and tensor isomorphism.

Talented Students Program, UNSW

Sydney, Australia

Academic Reading in Large Language Models with Quoc Le Gia

Term 1, 2025

- Completed supervised academic reading through the UNSW Talented Students Program, focusing on selected literature in large language models.
- Developed practice in reading research papers, extracting technical assumptions, and presenting mathematical and algorithmic ideas clearly.